

Exercise (0.2.3). Prove that if n is composite then there are integers a and b such that n divides ab but n does not divide either a or b .

Proof: Suppose that n is composite. By definition of composite there exists some positive divisors $a, b \neq 1, n$ such that $ab = n(1)$. Clearly $n \mid ab$. Note that $b = n(\frac{1}{a})$ and $a = n(\frac{1}{b})$. Since $\frac{1}{a}, \frac{1}{b} \notin \mathbb{Z}$ we have shown that $n \nmid a, b$. $\square$

Exercise (0.2.10). Prove for any given positive integer N there exist only finitely many integers n with $\varphi(n) = N$ where φ denotes Euler's φ -function. Conclude in particular that $\varphi(n)$ tends to infinity as n tends to infinity.

Proof: Let $N \in \mathbb{Z}^+$, and $X = \{n \in \mathbb{Z} : \varphi(n) = N\}$. By the Fundamental Theorem of Arithmetic, $n = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_s^{\alpha_s}$ for primes p_i and $\alpha_i \in \mathbb{Z}^+$. Consider prime p , the largest of primes p_i . It follows that,

$$N = \varphi(n) = \prod_{i=1}^s p_i^{\alpha_i-1} (p_i - 1) \geq (p - 1).$$

Therefore for all follows that $p_i \leq p \leq N + 1$.

Note that

$$N = \varphi(n) = \prod_{i=1}^s p_i^{\alpha_i-1} (p_i - 1) \geq \prod_{i=1}^s p_i^{\alpha_i-1} \geq \prod_{i=1}^s 2^{\alpha_i-1}.$$

Consider α , the largest element in the set of α_i . It follows that,

$$N \geq \prod_{i=1}^s 2^{\alpha_i-1} \geq 2^{\alpha-1}.$$

Therefore $\alpha_1 \leq \alpha \leq \lceil \log_2(N) + 1 \rceil$. Since p_i is finite, and s_i is finite, and whose elements are bounded above by some function of N then the set,

$$A = \{x = \prod_{i=1}^s q_i^{\beta_i} : q_i \in p_i, \beta_i \in \alpha_i\}$$

is finite with $a \in A$ bounded above by some function of N . Note that X is a subset of A and therefore X is finite, and for all $n \in X$, n is bounded above by some function of N . Thus as n tends to infinity $\varphi(n)$ also tends to infinity. $\square$

Exercise (0.3.9). Prove that the square of any odd integer always leaves a remainder of 1 when divided by 8.

Proof: Suppose $n \in \mathbb{Z}$ is odd. By the definition, for some $i \in \mathbb{Z}$, $n = 2(i) + 1$. Consider n^2 ,

$$n^2 = (2(i) + 1)(2(i) + 1) = 4(i)^2 + 4(i) + 1 = 4i(i + 1) + 1.$$

Note that when i is odd, $i + 1$ must be even and vice versa. Therefore we can always factor out a 2 from the product $i(i + 1)$ giving us for some $j \in \mathbb{Z}$,

$$sn^2 = 8(j) + 1.$$

□

Exercise (1.1.25). Prove that if $x^2 = \epsilon$ for all $x \in G$ then G is abelian.

Proof: Suppose $x^2 = \epsilon$ for all $x \in G$. With some algebra we get,

$$x^2 = \epsilon,$$

$$x^2(x^{-1}) = \epsilon(x^{-1}),$$

$$x = x^{-1}.$$

Now consider some $a, b \in G$, and with (4) of Proposition 1 we can consider the following,

$$ab = (ab)^{-1} = b^{-1}a^{-1} = ba.$$

Thus G is abelian.

□

Exercise (1.3.4b). Compute the order of the elements in S_4 .

Solution: First recall from the end of Section 1.3 that the order of a permutation is the l.c.m of the lengths of the cycles in its cycle decomposition. Now we can compute the order of each permutation in S_4 ,

Permutation (σ)	$ \sigma $
(1)	1
(12)	2
(13)	2
(14)	2
(23)	2
(24)	2
(34)	2

Permutation (σ)	$ \sigma $
(1234)	4
(1243)	4
(1324)	4
(1342)	4
(1423)	4
(1432)	4

Permutation (σ)	$ \sigma $
(123)	3
(124)	3
(132)	3
(134)	3
(142)	3
(143)	3
(234)	3
(243)	3

Permutation (σ)	$ \sigma $
(12)(34)	2
(13)(24)	2
(14)(23)	2

□